



Rajiv Gandhi University Of Knowledge Technologies - RK Valley

Andhra Pradesh

Accredited by 'NAAC' with 'B+' Grade

**(Established through Act No.18 of 2008 by the Govt. of Andhra Pradesh
and recognized as per Section 2(f), 12(B) of UGC Act, 1956)**

Department of Computer Science and Engineering
EaseCollaborate

Under The Guidance of
S.RAJESWARI

Presented by

Shaik Shareef (R200862)

Boggula Sai Anudeep (R200699)

Bojja Manjunath Reddy (R200510)

Agenda

1. Overview (Abstract & Introduction)
2. Problem & Objectives
3. System Analysis (Existing Systems)
4. Proposed Solution
5. Design & Architecture
6. Implementation & Features
7. Results
8. Conclusion & Future Scope

ABSTRACT

- EaseCollaborate is a full-stack web-based project management and collaboration platform
- Designed to help teams efficiently organize, track, and manage their work
- Uses a structured hierarchy of workspaces, projects, and tasks
- Supports role-based access control for secure and organized workflows
- Provides features like task assignment, progress tracking, comments, and dashboards
- Integrates AI-based summarization to generate insights
- Built using modern technologies: React, Node.js, Express, and MongoDB

Introduction

- Collaboration is essential in modern academic and professional environments
- Managing tasks across teams can be challenging without proper tools
- Traditional methods (manual tracking, spreadsheets) are inefficient
- Existing tools can be complex or not beginner-friendly
- EaseCollaborate provides a simple, structured, and centralized solution
- Helps teams manage projects, tasks, and collaboration effectively

Problem Statement

- Lack of a centralized platform for managing team tasks
- Difficulty in tracking progress across multiple users
- Miscommunication and lack of coordination in teams
- Existing tools are either too complex or require paid access
- Limited access to productivity insights in current systems

Objectives

- Develop a centralized platform for project and task management
- Enable efficient collaboration among team members
- Implement secure authentication and role-based access control
- Provide structured workflow using workspaces, projects, and tasks
- Integrate AI-based summarization for productivity insights
- Offer an intuitive and user-friendly interface

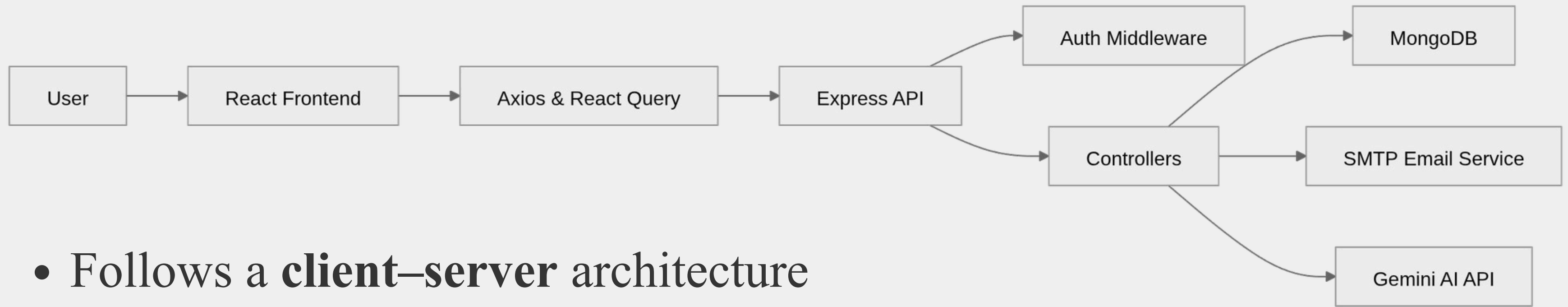
Existing Systems

- Trello : Simple and easy to use, but lacks advanced features and role control
- Jira : Powerful for development teams, but complex for beginners
- Asana : User-friendly interface, but many features are behind paid plans
- ClickUp : Feature-rich platform, but can be overwhelming due to complexity

Proposed System(EaseCollaborate)

- A centralized platform for managing projects and team collaboration
- Organizes work into Workspaces → Projects → Tasks
- Provides role-based access control for secure and structured workflows
- Enables task assignment, tracking, comments, and activity monitoring
- Offers dashboard analytics for better visibility of progress
- Integrates AI-based summarization for insights and productivity improvement
- Designed to be simple, efficient, and user-friendly

System Architecture



- Follows a **client–server** architecture
- **Frontend:** React + TypeScript (UI & user interaction)
- **Backend:** Node.js + Express (API & business logic)
- **Database:** MongoDB (data storage and management)
- **Communication:** REST APIs using Axios
- **Authentication:** JWT-based secure access
- **External Services:** Email (SMTP) and AI (Gemini API)

Modules

- **User Authentication & Role Management** : Handles login, registration, and secure access control
- **Workspace & Collaboration Management** : Manages teams, workspaces, and member roles
- **Project & Task Management** : Handles project creation, task assignment, and tracking
- **Dashboard & AI Summarization** : Provides analytics and AI-based productivity insights

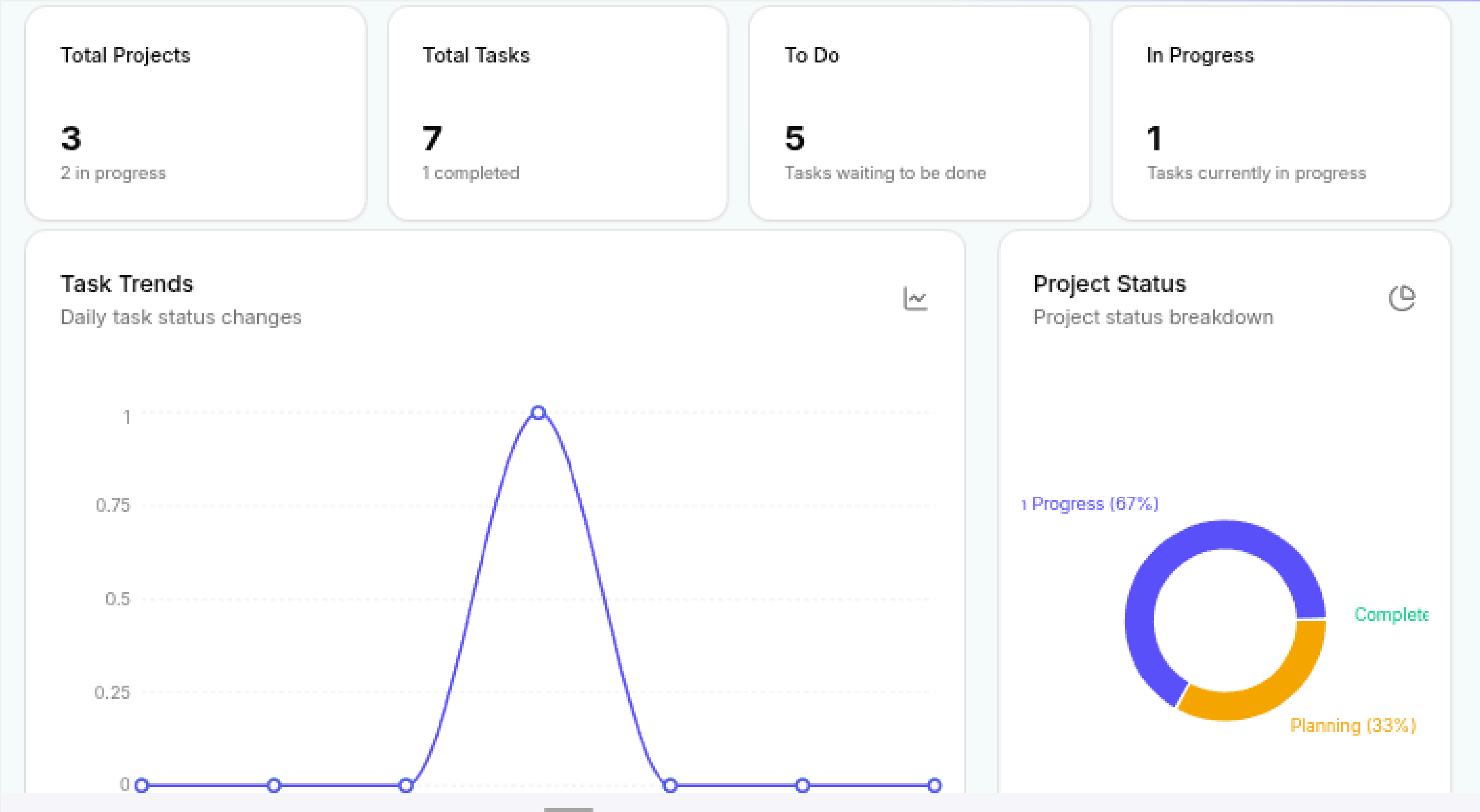
Implementation

- **Frontend** : React, TypeScript, React Query, Axios
- **Backend** : Node.js, Express, REST APIs
- **Database** : MongoDB with Mongoose
- **Authentication** : JWT-based secure login system
- **Validation & Security** : Zod validation, password hashing
- **AI Integration** : Gemini API for summarization

Key Features

- Secure user authentication and role-based access
- Workspace creation and team collaboration
- Project and task management with status tracking
- Task assignment, comments, and activity logs
- Dashboard analytics for progress monitoring
- AI-powered summarization for insights
- “My Tasks” view for personalized tracking

Results & Screenshots : Dashboard



WorkSpace

P

Product Team

 Invite

 Create Project

the workspace is related to product design

Members

D

A

M

Projects

Marketing Campaign
amnnagement

In Progress

Plan and track launch campaign activities.

Progress

0%

4 Tasks

 May 15, 2026

MobileApp Launch

Planning

Prepare the first release of the mobile application.

Progress

0%

4 Tasks

 May 5, 2026

Website Redesign

In Progress

Redesign the product website for better usability and branding.

Progress

0%

4 Tasks

 Jun 30, 2026

Project and Tasks

Marketing Campaign amnagement

Plan and track launch campaign activities.

Progress: 0%

Edit

Delete

Add Task

All Tasks

To Do

In Progress

Done

Status :

2 To Do

2 In Progress

0 Done

To Do

2

In Progress

2

Done

0

Low

🕒

✅

Review campaign performance metrics

D

A

M

📅 May 16, 2026

Medium

🕒

✅

Create social media content calender

A

M

D

📅 Apr 22, 2026

High

⚠️

✅

Write product launch announcement

D

A

M

📅 Apr 25, 2026

0 / 1 subtasks

Medium

⚠️

✅

Design email campaign template

A

M

📅 Apr 30, 2026

No tasks yet

Task Management

Write product launch announcement

 Watch

Archive

 Delete

High Priority

Status

In Progress



Write product launch announcement



Created at: about 1 hour ago

Description



Assignees

 Dhanush

 Ashraf

 Manjunath

3 selected

Priority

High



Sub Tasks



make office members participate

Watchers

No watchers

Activity



Ashraf added comment Sure on the work



Dhanush added comment Hey Ashraf can you convince office members to part...



Dhanush created subtask make office members participate

Future Scope

- Real-time updates using WebSockets
- Notification system for tasks and activities
- Mobile application development
- File and document sharing support
- Advanced analytics and reporting features
- Integration with third-party tools

Learning Outcomes & Conclusion

- Successfully developed a full-stack project management system
- Gained practical experience in React, Node.js, and MongoDB
- Implemented authentication and role-based access control
- Understood real-world system design and architecture
- Integrated AI features for productivity insights
- Improved problem-solving and development skills
- Demonstrated an efficient solution for team collaboration and task management

References

- React Documentation – <https://react.dev>
- Node.js Documentation – <https://nodejs.org>
- Express.js – <https://expressjs.com>
- MongoDB – <https://mongodb.com>
- Mongoose – <https://mongoosejs.com>
- Zod – <https://zod.dev>
- Gemini API – <https://ai.google.dev>

Thank You